

Linux Security Monitoring and Auditing Report

Assessment: Linux Security Monitoring and Auditing

Environment: Kali Linux [The authorized machine was down on the icdfa portal so i had to use kali linux]

Assessment Date: 1 October 2026

Assessment Framework: GRC102 Week 4 Practical Laboratory

Prepared by: Moyosore obasa

Organisation: IC DFA

1. Executive Summary

This assessment evaluated the security monitoring, logging and auditing capabilities of an authorised Kali Linux laboratory environment. The assessment focused on four key areas, auditd configuration and file monitoring, system and authentication log monitoring, security hardening using Lynis, and the governance of security-monitoring evidence.

The assessment successfully established and loaded custom auditd rules for monitoring sensitive account files, including `/etc/passwd` and `/etc/shadow`, as well as program execution. A controlled interaction with `/etc/passwd` generated an audit event, demonstrating that the monitoring rule was functioning.

System logs were subsequently reviewed using `journalctl`. The review identified evidence of privileged sudo activity and several system/service errors and warnings. These included kernel/graphics-related errors, systemd-related issues, and failures involving services such as `accounts-daemon` and Docker. The presence of these events does not, by itself, establish a security incident; however, they demonstrate the importance of continuous monitoring and appropriate investigation.

The assessment also established that Lynis version 3.1.6 was installed and an audit was initiated. The final Lynis hardening index, warnings and suggestions are pending capture and therefore have not been treated as confirmed findings in this report.

From a governance perspective, the assessment identified opportunities to strengthen privileged-activity accountability, monitoring of sensitive system files, and the review and remediation of system errors and warnings. These areas should be incorporated into ongoing security monitoring, evidence review and remediation processes.

2. Scope

The assessment covered the following areas specified in the practical laboratory:

- auditd installation, configuration and operation
- Monitoring of:
 - `/etc/passwd`
 - `/etc/shadow`
 - program execution
- Audit-event generation and review using `ausearch`
- Audit summaries using `aureport`
- System and security log review using `journalctl`
- Authentication and privileged activity monitoring
- Error and warning review
- Lynis security auditing
- Security monitoring and SIEM concepts
- Governance implications of security-monitoring evidence
- Control monitoring and remediation planning

3. Assessment Objectives

The assessment aimed to determine whether the Linux environment:

1. Generated auditable security events.
2. Monitored sensitive system files.
3. Recorded privileged activity.
4. Provided usable authentication and system logs.
5. Captured system errors and warnings.
6. Supported security hardening assessment.
7. Produced evidence suitable for security governance and audit.
8. Provided a foundation for centralised monitoring and SIEM integration.

4. Methodology

The assessment followed the practical laboratory methodology and consisted of the following activities:

4.1 Audit configuration

The `auditd` service was checked and configured with custom rules for:

`/etc/passwd`

`/etc/shadow`

`execve` system calls

The rules were loaded using `augenrules`, and their presence was verified using:

`sudo auditctl -l`

4.2 Controlled audit-event generation

A benign interaction with `/etc/passwd` was performed using:

`sudo nano /etc/passwd`

The file was exited without saving changes.

The resulting audit evidence was searched using:

`sudo ausearch -k passwd_changes`

4.3 Audit reporting

Audit summaries were reviewed using:

```
sudo aureport
```

```
sudo aureport --failed
```

```
sudo aureport --login
```

4.4 System-log review

The system journal was reviewed using:

```
sudo journalctl --since "today"
```

Additional filtering was performed for:

```
sudo journalctl --since "today" | grep -i "failed"
```

```
sudo journalctl --since "today" | grep -i "sudo"
```

```
sudo journalctl -p err --since "today"
```

```
sudo journalctl -p warning --since "today"
```

These activities align with the laboratory's required log-monitoring evidence.

4.5 Security hardening assessment

Lynis 3.1.6 was installed and the following audit was initiated:

```
sudo lynis audit system
```

The laboratory requires the hardening index, warnings, suggestions and report/log locations to be recorded.

5. Findings

1. Sensitive Account File Activity Was Successfully Audited

Risk: Medium

Status: Monitoring control operational

The configured `auditd` rule successfully captured activity involving `/etc/passwd`.

The audit record contained information including:

- `passwd_changes` audit key
- `/etc/passwd`
- Timestamp around 19:15 on 1 October 2026

- Effective UID associated with **root**
- An unset audit user ID (auid=4294967295)

Security significance

/etc/passwd is a sensitive account-related system file. Monitoring activity against it provides an audit trail that can support investigation and accountability.

However, the evidence does **not** establish that a particular human user intentionally modified the file. The unset auid means additional correlation may be required to establish human accountability.

Recommendation

Maintain monitoring of sensitive account files and correlate privileged audit events with authentication and session information where possible.

Remediation owner: Linux/System Administrator

Supporting owner: Security/GRC Team

2. Privileged Administrative Activity Is Being Logged

Risk: Medium

The journal contained multiple sudo records showing privileged commands executed with root privileges.

Examples included commands associated with:

- Checking auditd
- Installing auditd
- Starting and enabling auditd
- Restarting auditd

Security significance

Privileged activity represents a significant governance concern because administrative actions can alter security controls and system configurations.

The logs provide useful traceability by recording the user initiating the sudo operation and the command executed.

Recommendation

Implement periodic review of privileged activity and establish criteria for investigating unusual administrative commands.

Remediation owner: System Administrator

Control owner: Security/GRC Team

3. Multiple System and Service Errors Require Review

Risk: Medium

The journal contained multiple errors involving system and service components.

Examples included:

- vmwgfx errors related to the virtualisation/graphics environment
- systemd-ssh-generator termination
- systemd-journald journal-file issues
- accounts-daemon failures
- lightdm authentication/session-related errors
- Docker-related failures

Security significance

These events do not automatically indicate compromise. However, persistent service or logging failures could affect system availability or the reliability of security-monitoring evidence.

In particular, issues affecting logging services should receive attention because incomplete or unreliable logs can reduce the ability to investigate security events.

Recommendation

Investigate recurring errors, determine whether they are expected characteristics of the laboratory VM, and document the outcome.

Remediation owner: Linux/System Administrator

Supporting owner: Security/GRC Team

4. Lynis Audit Requires Completion of Evidence Capture

Risk: To be determined

Lynis version **3.1.6** was successfully installed and the security audit was initiated.

The captured output confirmed:

- Kali Linux environment
- Kernel 6.12.25
- Lynis 3.1.6
- Log: /var/log/lynis.log
- Report: /var/log/lynis-report.dat

6. Risk Summary

Finding	Risk
Sensitive account file activity monitoring	Medium
Privileged activity monitoring/review	Medium
System/service errors and warnings	Medium

7. Recommendations

1. Maintain sensitive-file auditing

Continue monitoring `/etc/passwd` and `/etc/shadow` through `auditd`.

Ensure audit rules are loaded after system changes and periodically verify that the monitoring control remains active.

2. Strengthen privileged-activity review

Establish a defined review process for `sudo` and other privileged activity.

Reviews should consider:

- User
- Timestamp
- Command
- Business/administrative justification
- Related system events

3. Establish an error-management process

System and security errors should be assigned to an appropriate owner and tracked through investigation and resolution.

Repeated errors should be distinguished from expected laboratory/environmental conditions.

4. Consider SIEM integration

The audit and journal data could be forwarded to a SIEM to support:

- Correlation
- Alerting
- Investigation
- Incident tracking

- Management reporting

8. Remediation Ownership

Remediation	Primary Owner	Evidence closure
Verify sensitive-file audit rules remain active	System Administrator	<code>auditctl -l</code> + successful <code>ausearch</code>
Establish periodic privileged-activity review	Security Team	Reviewed <code>sudo</code> log + review record
Investigate recurring system/service errors	System Administrator	Investigation record

Conclusion

The assessment demonstrated that the Kali Linux environment has functioning security-monitoring capabilities through `auditd` and `journalctl`. Sensitive account-file activity was successfully captured, while journal records provided evidence of privileged administrative activity and various system/service conditions.

The assessment also demonstrated an important governance principle: collecting logs is not sufficient by itself. Security evidence must be attributable, reviewed, protected and linked to an accountable owner and defined control objective.